

Configuration Automation using Ansible

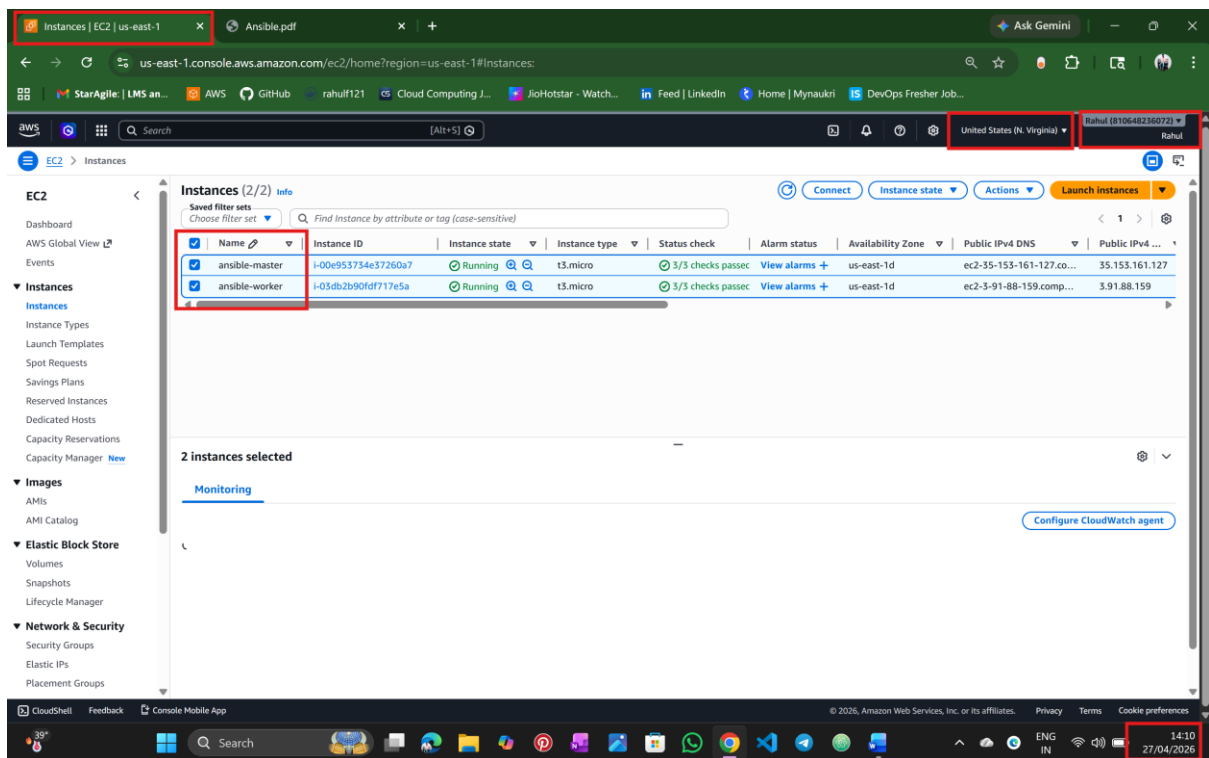
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L2 - Create and Execute Ansible Playbook to Install Docker and Run the Docker Application Image created in Docker Module

Step:1 login into AWS console and create ec2 instance



Step:2 This is ansible master node

The screenshot shows the AWS Management Console for an EC2 Instance Connect session. The browser tabs at the top include 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and another 'EC2 Instance Connect | us-east-1'. The address bar shows the URL: `us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-00e953734e37260a7&osUs...`. The AWS account name 'Rahul' and ID '810648236072' are visible in the top right. The terminal window displays the following output:

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Sun Apr 26 08:58:45 2026 from 18.206.107.20
[ec2-user@ip-172-31-47-209 ~]$ sudo su
[root@ip-172-31-47-209 ec2-user]# ansible --version
ansible [core 2.15.3]
  config file = None
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.9/site-packages/ansible
  ansible collection location = /root/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.9.25 (main, Mar 13 2026, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3.9)
  jinja version = 3.1.4
  libyaml = True
[root@ip-172-31-47-209 ec2-user]#
```

Below the terminal, the instance details for 'i-00e953734e37260a7 (ansible-master)' are shown, including Public IP: 35.153.161.127 and Private IP: 172.31.47.209. The bottom status bar shows the time as 14:13 on 27/04/2026.

Step:3 This is ansible worker node

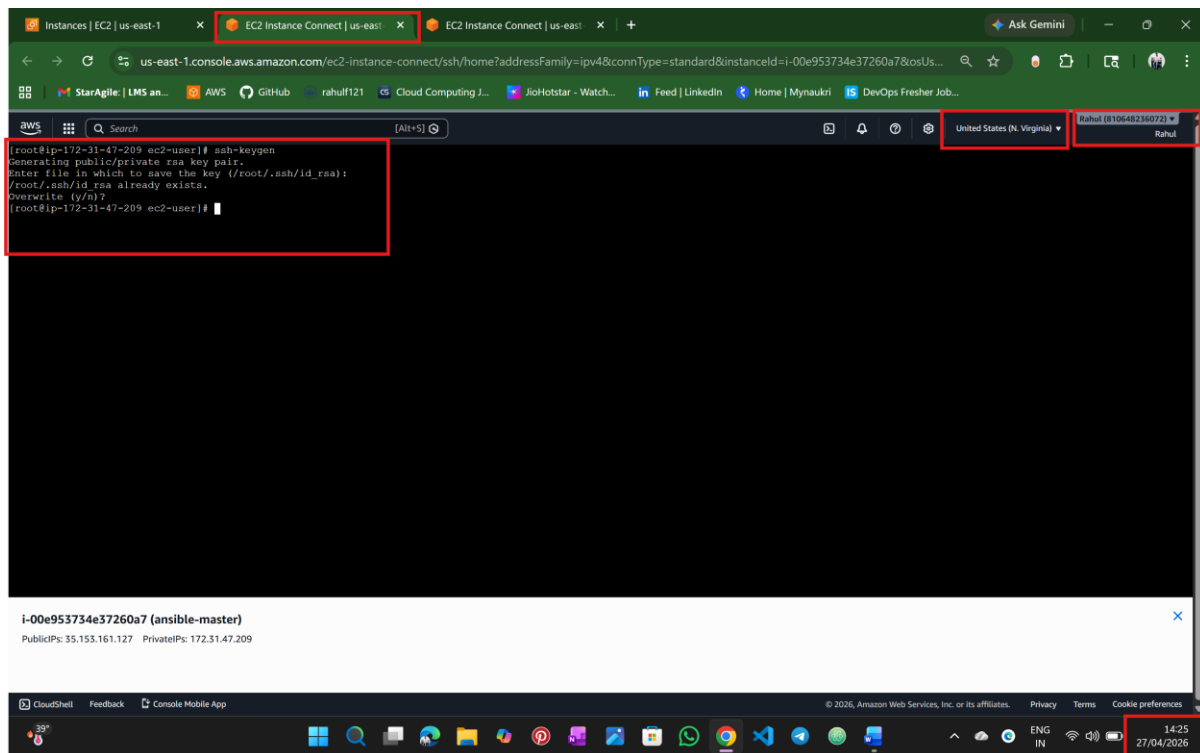
The screenshot shows the AWS Management Console for an EC2 Instance Connect session. The browser tabs at the top include 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and another 'EC2 Instance Connect | us-east-1'. The address bar shows the URL: `us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-03db2b90df717e5a&osUser=...`. The AWS account name 'Rahul' and ID '810648236072' are visible in the top right. The terminal window displays the following output:

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Sun Apr 26 10:09:18 2026 from 172.31.47.209
[ec2-user@ip-172-31-38-210 ~]$ sudo su
[root@ip-172-31-38-210 ec2-user]# ansible --version
bash: ansible: command not found
[root@ip-172-31-38-210 ec2-user]#
```

Below the terminal, the instance details for 'i-03db2b90df717e5a (ansible-worker)' are shown, including Public IP: 3.91.88.159 and Private IP: 172.31.38.210. The bottom status bar shows the time as 14:13 on 27/04/2026.

Step:4 connect ansible master node and worker node with ssh-keygen, here I already connected to it.

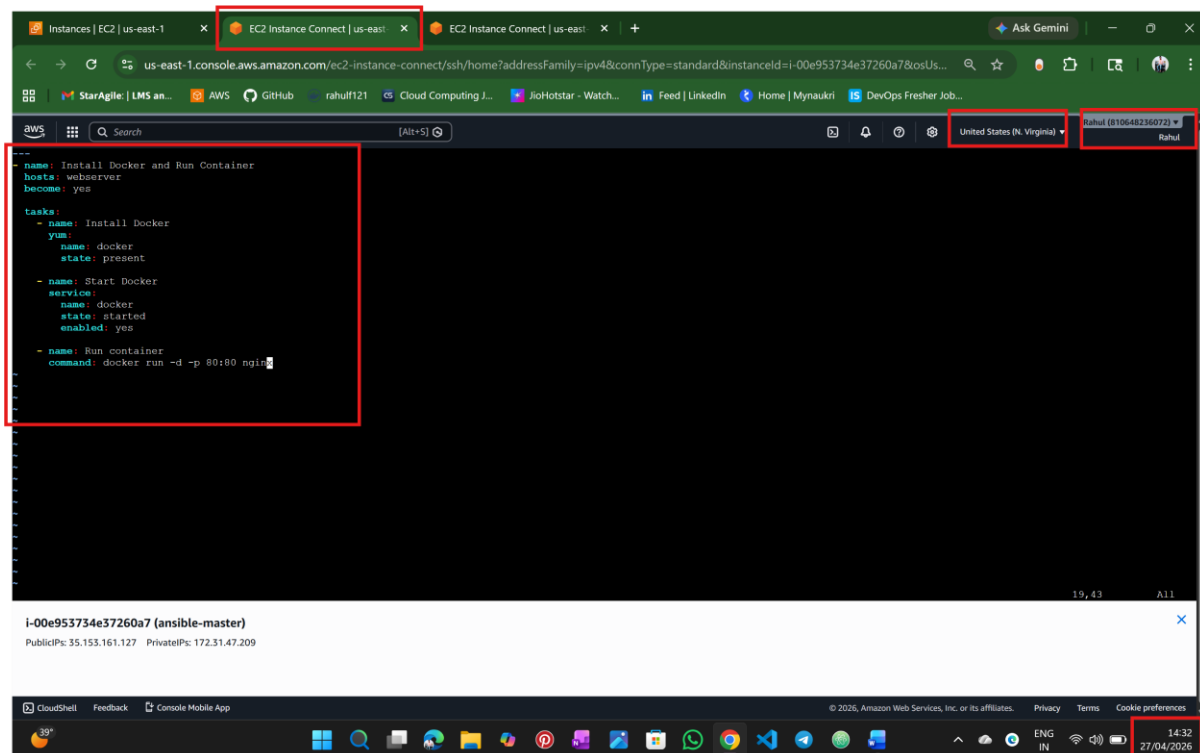


The screenshot shows the AWS CloudShell interface. The terminal window displays the following output for the `ssh-keygen` command:

```
[root@ip-172-31-47-209 ec2-user]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
/root/.ssh/id_rsa already exists.
Overwrite (y/n)?
```

The terminal prompt is `[root@ip-172-31-47-209 ec2-user]#`. The AWS console header shows the region as "United States (N. Virginia)" and the user as "Rahul". The instance details at the bottom indicate the instance is `i-00e953734e37260a7 (ansible-master)` with public IP `35.153.161.127` and private IP `172.31.47.209`. The system clock at the bottom right shows 14:25 on 27/04/2026.

Step:5 This is yaml file



The screenshot shows the AWS CloudShell interface with an Ansible playbook pasted into the terminal. The playbook content is as follows:

```
---
- name: Install Docker and Run Container
  hosts: webserver
  become: yes
  tasks:
    - name: Install Docker
      yum:
        name: docker
        state: present
    - name: Start Docker
      service:
        name: docker
        state: started
        enabled: yes
    - name: Run container
      command: docker run -d -p 80:80 nginx
```

The terminal prompt is `[root@ip-172-31-47-209 ec2-user]#`. The AWS console header shows the region as "United States (N. Virginia)" and the user as "Rahul". The instance details at the bottom indicate the instance is `i-00e953734e37260a7 (ansible-master)` with public IP `35.153.161.127` and private IP `172.31.47.209`. The system clock at the bottom right shows 14:32 on 27/04/2026.

Step:6 verify and check the connect and run the yaml file

The screenshot shows the AWS Management Console interface for an EC2 instance. The terminal window displays the following commands and output:

```
[root@ip-172-31-47-209 /]# vi hosts
[root@ip-172-31-47-209 /]# ansible -i hosts webserver -m ping
[WARNING]: Platform linux on host 172.31.38.210 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.38.210 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}

[root@ip-172-31-47-209 /]# vi docker-playbook.yml
[root@ip-172-31-47-209 /]# ansible-playbook -i hosts docker-playbook.yml

PLAY [Install Docker and Run Container] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 172.31.38.210 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.38.210]

TASK [Install Docker] *****
changed: [172.31.38.210]

TASK [Start Docker] *****
changed: [172.31.38.210]

TASK [Run container] *****
changed: [172.31.38.210]

PLAY RECAP *****
172.31.38.210      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

[root@ip-172-31-47-209 /]#
```

Below the terminal output, the instance details for **i-00e953734e37260a7 (ansible-master)** are shown, including Public IP: 35.153.161.127 and Private IP: 172.31.47.209.

Step:7 verify the container

8

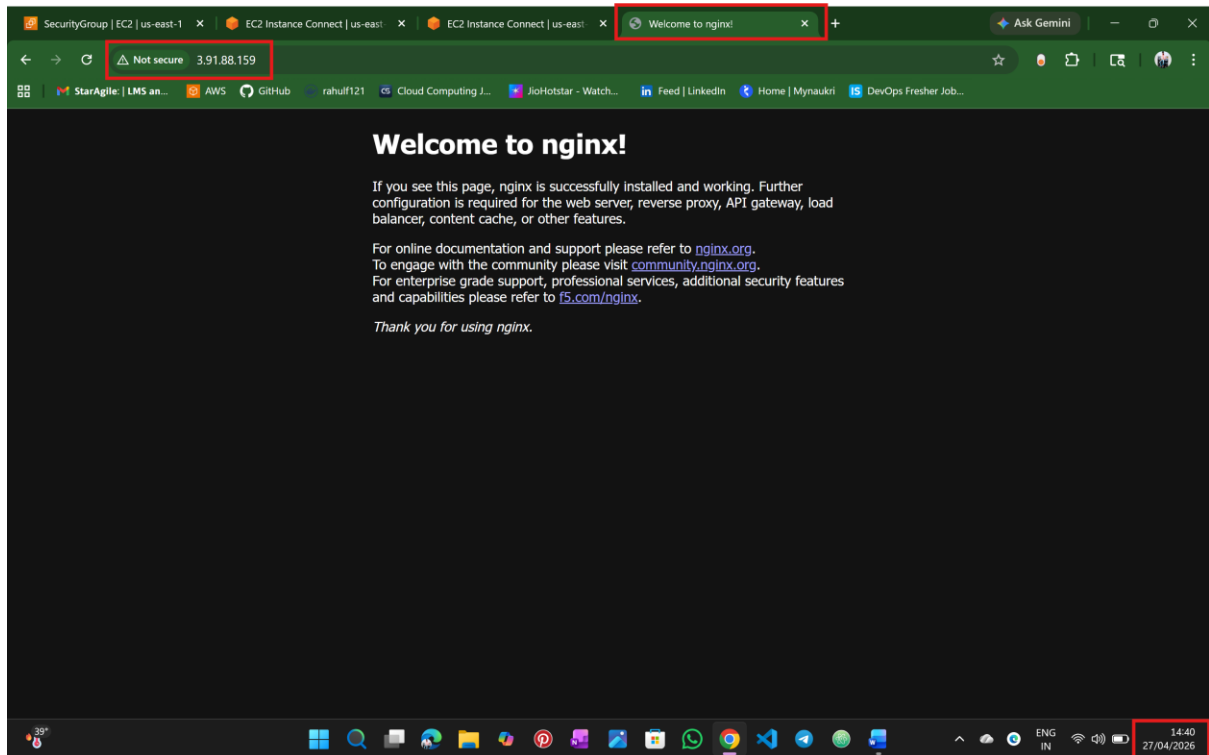
The screenshot shows the AWS Management Console interface for an EC2 instance. The terminal window displays the following commands and output:

```
[ec2-user@ip-172-31-38-210 ~]$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
654d249cc84e   nginx    "/docker-entrypoint..." 3 minutes ago    Up 3 minutes    0.0.0.0:80->80/tcp, :::80->80/tcp    condescending_varahamihira

[ec2-user@ip-172-31-38-210 ~]$ sudo usermod -s /usr/sbin/docker ec2-user
[ec2-user@ip-172-31-38-210 ~]$
```

Below the terminal output, the instance details for **i-03db2b90fdf717e5a (ansible-worker)** are shown, including Public IP: 3.91.88.159 and Private IP: 172.31.38.210.

Step:8 access the website using worker node ip address



Nalathati Rahul